

ALINDA BHATTACHARJEE

BIM Engineer | Construction Automation | UAV, LiDAR & Photogrammetry | Machine Learning in AEC

Dhaka, Bangladesh

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Profile

BIM Engineer and construction technology researcher with experience in UAV photogrammetry, point-cloud processing, BIM-based progress monitoring, and AI-driven construction applications. Currently working on Revit-based HVAC/MEP design, prefabrication, and BIM automation. Interested in LLMs, digital twins, and DfMA to develop intelligent and automated construction workflows.

Academic Credentials

Bachelor of Science in Building Engineering and Construction Management

January 2019 – April 2024

Khulna University of Engineering & Technology (KUET)

CGPA: 3.08 (Out of 4)

Major: Construction Engineering and Management

Thesis: Real-Time Construction Progress Monitoring: Integrating UAV, Photogrammetry and BIM Technologies

Publications

Peer-Reviewed Journal Article

- **Bhattacharjee, A.**, Rana, M. J., Zaman, A. A. U., & Dang, H. (2026). Integrated information system for periodic construction progress monitoring: Utilizing UAV, photogrammetry, and BIM for enhanced decision-making and optimization. *Civil Engineering Sciences*, 2, Article 0022. doi:10.34133/cesci.0022

Peer-Reviewed Conference Proceedings

- **Bhattacharjee, A.**, Ahmed, E., & Rana, M. J. (2025). Automation in construction quality control: Integrating Unmanned Aerial Vehicle (UAV), CNNs, and photogrammetry for structural element detection and measurement precision. *Advances in Engineering Research*, 181–189. doi:10.2991/978-94-6463-884-4_22
- Ahmed, E., **Bhattacharjee, A.**, & Anzum, A. (2025). Reducing false alarms in buildings with UAV-based fire detection and monitoring through a cloud-integrated approach. *Advances in Engineering Research*, 120–127. doi:10.2991/978-94-6463-884-4_15

Research Interests

BIM Automation, Construction Progress Monitoring, UAV, LiDAR and Photogrammetry, Point Clouds, Computer Vision for Construction, Digital Twins, Prefabrication and DfMA, Machine Learning in AEC

Research Experience

Undergraduate Thesis: UAV, Photogrammetry and BIM for Construction Progress Monitoring

KUET

- Monitored construction progress of the Foreign Student's Dormitory and Visiting Faculty Accommodation at KUET through UAV (drone) operation and photogrammetric analysis.
- Generated point clouds from UAV imagery to support BIM-integrated progress monitoring.

Seminar Project: Application of BIM in Building Energy Performance Simulation (BEPS)

KUET

Professional Experience

BIM Engineer & Prefabrication Automation Specialist

January 2025 – Present

HawarIT, Bangladesh

www.hawarit.com

- Delivered prefabricated design and development for 7+ multistoried residential and commercial apartment projects in compliance with Dutch [NEN](#) and [ISSO](#) standards, including HVAC and plumbing prefabrication drawings for a 23-story residential building.
- Produced 50+ complete HVAC and plumbing installation models for single residential buildings, compliant with NEN and ISSO; performed 200+ HVAC heat-load calculations and MEP coordination.
- Used Dynamo to automate prefabrication assemblies for plumbing and ventilation systems and to generate RFA families for complete apartment plumbing assemblies with accurate connector placement.
- Developed visual programming workflows to verify pipe-to-reducer connections and Muff (MOF) spigot connections in plumbing and ventilation systems.

Software and Technical Skills

BIM & MEP	Autodesk Revit (HVAC/MEP, prefabrication and installation modeling), Dynamo, Navisworks, Autodesk Construction Cloud, Fusion 360, Autodesk Design Review
Reality Capture	UAV operation, DroneDeploy, RealityCapture, photogrammetry, point clouds
Programming	Python, C++
ML Libraries	scikit-learn, PyTorch, NumPy, pandas, matplotlib
ML Methods	Regression, tree-based and boosting methods, SVM, clustering, DNN, CNN, RNN, Transformers
Engineering	AutoCAD, ETABS, Autodesk Robot Structural Analysis
Management	Primavera P6, MS Project
Visualization	SketchUp, Enscape, Lumion

Training

Applied Machine Learning	BUET • IAC, Batch 30	4.00 / 4.00 (91%)
Computer Science Fundamentals	Phitron	Fall 2024

Applied Machine Learning coverage: supervised and unsupervised learning, deep learning (DNN, CNN, RNN, Transformers), model evaluation (cross-validation, ROC-AUC, precision/recall), feature engineering and data preprocessing.

Academic Projects

- *Structural Analysis of a Twelve-Story Residential Building:* Structural modeling and analysis using ETABS.
- *Project Planning and Scheduling:* WBS, scheduling, and resource allocation with project-duration optimization in Primavera P6.
- *Residential Building Design and Estimation:* Complete working drawings for a five-story residential building; material cost estimate for a six-story residential building (group projects).

Leadership Experience

Vice President	May 2023 – April 2024
BECM Technical Club, KUET	Supported technical and educational initiatives, organized training sessions, and mentored junior students.
Program Coordinator	June 2022 – May 2023
BECM Technical Club, KUET	Organized and conducted training sessions on Revit, AutoCAD, and ETABS for junior students.
Organizing Secretary	September 2023 – May 2024
BUILTECH FEST 5.0, KUET	Coordinated advertising, promotional content, drone videography, and event documentation.

Linguistic Proficiency

- Bengali (Native Proficiency)
- English (Excellent reading and writing; good speaking)

References

Name	Md. Jewel Rana	Ahmad Akib Uz Zaman
Designation	Assistant Professor (on study leave), Dept. of BECM; Graduate Research Assistant, Washington State University	Engineer-I
Institution	Khulna University of Engineering & Technology	Wyoming Department of Transportation (WYDOT), 680 Fort St., Buffalo, WY 82834, USA
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